

KRISHNA Lab Guide

Gopalkrishna Advanced Rural Research Foundation (GARRF) • Virtual Semiconductor Chip Design

What Students Will Learn

Through this interactive platform, students transition from abstract theory into an active, hands-on understanding of modern silicon engineering. The platform bridges the gap between physics, materials science, and industrial factory management.

Core Philosophy: Engineering is about balancing trade-offs. Students learn that improving one metric (e.g., clock speed) often degrades another (e.g., thermal stability), requiring precise optimization.

1. End-to-End Silicon Fabrication Flow

Students gain operational understanding of how sand is turned into ultra-dense microchips across 6 core stages:

- **Design:** Micro-architectural decisions dictate clock speeds, gate density, and thermal design power (TDP).
- **Lithography:** How light exposure transfers mask patterns onto photoresist, and the impact of misalignment on critical dimensions.
- **Deposition:** Building microscopic oxide layers and metallic interconnect lines through chemical vapor deposition.
- **Etching:** Using directional plasma to carve trench features and transistor gates without damaging the silicon lattice.
- **Doping:** Ion implantation to alter silicon electrical properties to form p-n junctions.
- **Metrology:** Automated laser scanning and defect mapping to ensure quality control.

2. Multi-Scale Digital Twin Modeling

Students manage semiconductor manufacturing across four distinct engineering levels:

Twin Level	Engineering Focus & Learning Outcomes
Chip Twin	Transistor physics, gate delays, clock frequency dynamics, thermal hot-spots, and voltage trade-offs.
Wafer Twin	Spatial yield engineering, edge-roll-off effects, defect density formulas, and whole-wafer profitability.
Equipment Twin	Plasma physics, vacuum chamber dynamics, gas balances, and temperature control in fabrication machines.

Step-by-Step Guide to Using the HTML Lab

The interactive web application is designed to simulate a real-world foundry control dashboard. Follow these steps to maximize your learning experience:

- **Step 1: Launch the Lab**

Open the `index.html` file in any modern web browser. The interface is split into a Control Panel on the left and a Visualization Canvas on the right.

- **Step 2: Explore the Digital Twins**

Use the top navigation tabs in the Control Panel to switch between the **Chip**, **Wafer**, **Equipment**, and **Fab** twins. Observe how the central graphic updates to reflect the selected scale. Notice the micro-engraved *"Gopalkrishna Advanced Rural Research Foundation"* text around the central chip die.

- **Step 3: Run the 6-Stage Process**

In the Fab or Equipment view, walk through the 6 stages (Design, Litho, Deposition, Etch, Doping, Metrology). Click the buttons sequentially to see how raw silicon transforms into a patterned wafer.

- **Step 4: Adjust Live Parameters**

Locate the sliders for Voltage, Temperature, and Pressure. Move these sliders and watch the real-time telemetry graphs (Yield %, Temperature °C, Frequency GHz) respond.

Experiment: Try pushing the temperature too high to see how thermal throttling affects the simulated chip frequency and wafer yield.

- **Step 5: Utilize the Help System**

Click the **"HELP & GUIDANCE"** button located in the top navigation bar at any time to review learning objectives or download this PDF manual.

AI Assistant Integration: While adjusting parameters, think about the physical reasons behind the metrics changing. The simulated environment represents the exact non-linear physics encountered in modern 5nm/3nm fabrication nodes.

Conclusion

By completing this virtual lab, you have simulated the responsibilities of a Process Engineer, Yield Engineer, and Fab Manager. You are now equipped with a systemic understanding of the world's most complex manufacturing process.